Photoelectric effect and determination of Planck's constant

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1 Introduction

When a metal is subject to electromagnetic radiation, the electrons in the metal may be excited by incident photons to the point that they leave the metal in a phenomenon known as the *photoelectric effect*, which was first explained theoretically in the first of Einstein's 1905 *annus mirabilis* papers. This explanation is bound up in the history of another phenomenon, *blackbody radiation*, of which the investigation by Max Planck first introduced the notion that energy may be quantized (though Planck was hesitant to assign physical significance to their findings).

1.1 Blackbody radiation

Everything with a temperature above absolute zero, as it turns out, emits thermal electromagnetic radiation. This is most readily observed in things like red-hot metal, thermal imaging cameras, and sunlight.

1.1.1 Rayleigh and Jeans

Einstein built off the results of Planck, who in turn had corrected the results of Jeans and Rayleigh.

Rayleigh and Jeans sought to model the behavior of a *blackbody*, a hypothetical object that absorbs all incident light, which they could approximate with a *Jeans cube*, a hollow, metal cube with a small hole, such that the hole functions as a blackbody.

One dimension Such a cube, if heated, would give rise to standing electromagnetic waves along the cube's 3 axes, as the light emitted from the walls would reflect back onto the incoming wave with a 180° phase shift. Such a standing wave, in one dimension, is described by

$$E(x,t) = 2E_0 \sin(kx)\cos(\omega t)$$

for a wavenumber $k = 2\pi/\lambda$.

But these waves are subject to the boundary condition that $E = 0$ at the walls, as any electric field there would immediately be cancelled by the movement of the charge of the wall itself. For a cube of side length L, this implies

$$\begin{aligned}
&sin(kL) = 0 \\
&\implies kL = n\pi \\
&\implies \frac{n\pi}{L} = \frac{2\pi}{\lambda} \\
&\implies \lambda = \frac{2L}{n} \qquad \forall n \in \mathbb{N},
\end{aligned}$$

or in terms of frequency,

$$\nu = \frac{nc}{2L}.$$

This result limits the frequencies of light possible, at which point it makes sense to talk about the *density of states*

$$\rho = 2\frac{n}{\nu} = \frac{4L}{c},$$

such that $\rho\Delta\nu$ is the number of states in a range of frequencies, and where the factor of two accounts for the polarization of the wave.

This result is only one dimensional. It must now be generalized to three dimensions.

Three dimensions Radiation of some frequency ν inside the cube would be decomposable into three components along each axis of the cube, subject to previous results, that is,

$$\begin{aligned}
&\mathbf{E}(\mathbf{r}, t) = \begin{pmatrix} E_x(x,t) \\ E_y(y,t) \\ E_z(z,t) \end{pmatrix} = \begin{pmatrix} 2E_{0x}\sin(2\pi x/\lambda_x)\cos(\omega t) \\ 2E_{0y}\sin(2\pi y/\lambda_y)\cos(\omega t) \\ 2E_{0z}\sin(2\pi z/\lambda_z)\cos(\omega t) \end{pmatrix} \\
&\implies \begin{cases} \lambda_x = 2L/n_x \\ \lambda_y = 2L/n_y \\ \lambda_z = 2L/n_z \end{cases} \\
&\implies \lambda = \sqrt{\lambda_x^2 + \lambda_y^2 + \lambda_z^2} \\
&\implies \nu = \frac{c}{\lambda} = \frac{c}{2L}\sqrt{n_x^2 + n_y^2 + n_z^2}.
\end{aligned}$$

Rayleigh and Jeans proceeded by calculating the number of states per range of frequencies, and then applying a result from statistical mechanics in order to arrive at the average total energy.

In the one dimensional case, this number of states was calculable by multiplying the density of states, which was constant and easily obtained, by the length of the range of frequencies. In the three-dimensional case, it is not immediately apparent what this density is.

A geometric approach yields the answer. By treating the numbers n_x, n_y, and n_z as describing points in $\mathbb{N}^3$, it makes sense to treat the quantity $\sqrt{n_x^2 + n_y^2 + n_z^2}$ as a radius. This means that, for some frequency ν, the set of points in $\mathbb{N}^3$ that lie on a sphere of radius $n = \frac{2L}{c}\nu$ describes all the possible combinations of n_x, n_y, and n_z for that frequency.

This is not in itself a useful result, as the intersection of such a sphere and $\mathbb{N}^3$ is empty for most frequencies. Instead, a range of frequencies $\mathrm{d}\nu$ must be considered, and therefore the intersection of a shell of thickness $\mathrm{d}r = \frac{2L}{c}\mathrm{d}\nu$ with $\mathbb{N}^3$. The size of this intersection must be equal to the volume of the part of the shell that lies in $\mathbb{R}^3_+$ times the volumetric density of points in $\mathbb{N}^3$, which may be seen trivially to equal 1:

$$\begin{aligned}
\mathrm{d}N(r) &= \rho_{\mathbb{N}}\mathrm{d}V(r) = \frac{1}{8}4\pi r^2\mathrm{d}r \\
= \mathrm{d}N(\nu) &= \frac{1}{2}\pi\left(\frac{2L}{c}\right)^3\nu^2\mathrm{d}\nu \\
\implies 2\frac{\mathrm{d}N}{\mathrm{d}\nu} &= \frac{8\pi L^3}{c^3}\nu^2 = \rho(\nu).
\end{aligned}$$

Energy density Rayleigh and Jeans wanted ultimately to derive an expression for spectral energy density (energy per unit volume per unit wavelength), as that was the quantity that could be measured empirically for an actual Jeans cube with which they wanted to compare their model.

To this end, the pair treated the system as a classical thermodynamic system, such that the probability of each state follows the Boltzmann distribution and the average energy per state may be expressed

$$\begin{aligned}
\bar{E} &= \sum \text{energy} \times \text{probability} \\
&= \sum_i \varepsilon_i p_i \\
&= \sum_i \varepsilon_i \left(Z^{-1}\exp\left(-\frac{\varepsilon_i}{k_\mathrm{B}T}\right)\right) \\
&= \frac{\sum_i \varepsilon_i \exp\left(-\frac{\varepsilon_i}{k_\mathrm{B}T}\right)}{\sum_j \exp\left(-\frac{\varepsilon_j}{k_\mathrm{B}T}\right)},
\end{aligned} \tag{1}$$

at which point the assumption is made in classical physics that energy levels are continuous, that is, energy may take on any real value. Here, a justification can be made (as is often) to turn the summations into integrals with respect to E:

$$\begin{aligned}
\bar{E} &= \frac{\int_0^\infty E\exp\left(-\frac{E}{k_\mathrm{B}T}\right)\mathrm{d}E}{\int_0^\infty \exp\left(-\frac{E}{k_\mathrm{B}T}\right)\mathrm{d}E} \\
&= \frac{(k_\mathrm{B}T)^2}{k_\mathrm{B}T} = k_\mathrm{B}T,
\end{aligned}$$

which is just the result of the equipartition theorem from statistical mechanics.

Spectral energy density may now be given as

$$\begin{aligned} u_T(\nu) &= \frac{\text{energy}}{\text{state}} \times \frac{\text{states}}{\text{unit wavelength}} / \text{volume} \\ &= \bar{E}\rho(\nu)/L^3 \\ &= \frac{8\pi k_\mathrm{B} T}{c^3}\nu^2, \end{aligned} \tag{2}$$

which is the *Rayleigh-Jeans law*.

Though this result agreed well with data at low frequencies, it predicted spectral energy density to go to infinity with increasing frequency, which was completely unphysical and at odds with empirical data and the theoretical results of Boltzmann as given by the Stefan-Boltzmann law.

1.1.2 Wien

The work of Wilhelm Wien was also instrumental in the formulation of Planck's law.

1.1.3 Planck

At this point, there existed two laws that were valid at opposite ends of the electromagnetic spectrum.

Max Planck sought to correct the results of Rayleigh and Jeans, by first recognizing that, in the limit as $\nu \to 0$, $\bar{E} \to k_\mathrm{B}T$, which implied that $\bar{E}$ was in fact a function of ν.

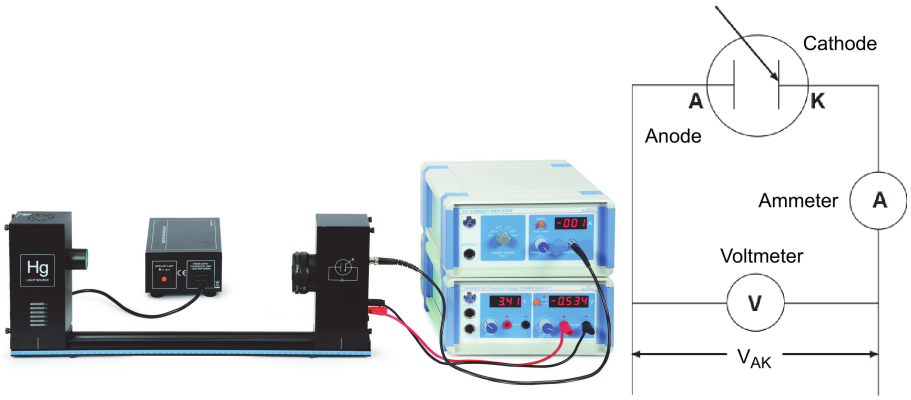
1.2 Photoelectric effect

2 Materials and Procedure

3 Results

4 Discussion

5 Conclusions



(a) Experimental setup (b) Circuit diagram

Figure 1